

# Zeros, Multiplicity, and Graph Behavior

## Answer Key

Algebra 2   Polynomial Factoring, Roots, and Graphs

### Quick Answers

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1.  $x(x-2)(x+2)$

2.  $-1, 3$

3.  $x(x-5)(x+2)$

4.  $-\infty, \infty$

5.  $0, 4$

6.  $x^3 - 3x + 2$

7.  $-2, -1, 1, 2$

8.  $\infty, \infty$

9.  $x^3 - x^2 - 8x + 12$

10.  $-2, 0, 3$ 

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### Curriculum

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**Level:** Algebra 2   Polynomial Factoring, Roots, and Graphs

HSA-APR.B.3 – *problems 1, 2, 3, 4, 5, 6, 7, 8, 9, 10*

*Difficulty 1–5 of 5 across 13 tagged checks.*

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**1.** Factor  $f(x) = x^3 - 4x$  and list its zeros.

Pull the greatest common factor first — skipping it is what loses a zero:  $x^3 - 4x = x(x^2 - 4)$ .

The second factor is a difference of squares,  $x^2 - 4 = (x - 2)(x + 2)$ .  $x(x - 2)(x + 2)$

Each factor is linear and appears once, so the zeros are  $x = 0$ ,  $x = 2$ ,  $x = -2$ , each of multiplicity 1: the graph crosses the axis at all three.

**2.**  $g(x) = (x + 1)(x - 3)^2$ : zeros, multiplicity, and behavior.

The polynomial is already factored, so read the zeros straight off: setting each factor to zero gives  $x = -1$  from  $(x + 1)$  and  $x = 3$  from  $(x - 3)^2$ .

$x = -1$  has multiplicity 1 (odd) — the graph *crosses* there.  $x = 3$  has multiplicity 2 (even) — the graph *touches* the axis and turns back, without changing sign.  $x = -1, x = 3$

Degree check:  $1 + 2 = 3$ , so  $g$  is cubic with a positive leading coefficient.

**3.** Factor  $h(x) = x^3 - 3x^2 - 10x$  and list its zeros.

GCF first:  $x^3 - 3x^2 - 10x = x(x^2 - 3x - 10)$ . Then factor the trinomial — two numbers multiplying to  $-10$  and adding to  $-3$  are  $-5$  and  $2$ :  $x^2 - 3x - 10 = (x - 5)(x + 2)$ .

$x(x - 5)(x + 2)$

Zeros  $x = 0$ ,  $x = 5$ ,  $x = -2$ , all of multiplicity 1, so the graph crosses at each.

**4.** End behavior of  $f(x) = -2x^3 + 5x - 1$ .

Only the leading term matters as  $|x|$  grows, because  $-2x^3$  outgrows  $5x$  and the constant. The degree is 3 (odd), so the two ends go opposite ways, and the leading coefficient  $-2$  is negative, so the right end falls.

$$\text{as } x \rightarrow \infty, f(x) \rightarrow \boxed{-\infty} \qquad \text{as } x \rightarrow -\infty, f(x) \rightarrow \boxed{+\infty}$$

Expected sentence: “The odd degree makes the ends opposite; the negative leading coefficient decides which end goes down.”

**5.**  $p(x) = x^2(x - 4)^3$ : degree, zeros, multiplicity, behavior.

Adding the exponents gives the degree:  $2 + 3 = 5$ , so  $p$  is quintic with a positive leading coefficient.

$x = 0$  has multiplicity 2 (even) — the graph *touches* the axis and turns back.  $x = 4$  has multiplicity 3 (odd and greater than one) — the graph *crosses*, but flattens against the axis as it passes.

$$\boxed{x = 0, x = 4}$$

**6.** Least-degree polynomial crossing at  $x = -2$  and bouncing at  $x = 1$ .

Crossing means an odd multiplicity, and the least odd multiplicity is 1, so  $(x + 2)$  appears once. Bouncing means an even multiplicity, least is 2, so  $(x - 1)^2$ . With leading coefficient 1:

$$(x + 2)(x - 1)^2 = (x + 2)(x^2 - 2x + 1) = x^3 - 2x^2 + x + 2x^2 - 4x + 2$$

Collecting like terms gives the expanded form.

$$\boxed{x^3 - 3x + 2}$$

Check: substituting  $x = 1$  gives  $1 - 3 + 2 = 0$  ✓ and  $x = -2$  gives  $-8 + 6 + 2 = 0$  ✓

**7.** Solve  $x^4 - 5x^2 + 4 = 0$ .

Treat it as a quadratic in  $x^2$ : two numbers multiplying to 4 and adding to  $-5$  are  $-1$  and  $-4$ , so  $x^4 - 5x^2 + 4 = (x^2 - 1)(x^2 - 4)$ . Each factor is a difference of squares:

$$(x^2 - 1)(x^2 - 4) = (x - 1)(x + 1)(x - 2)(x + 2)$$

Four distinct linear factors, each of multiplicity 1, so the graph crosses at all four.

$$\boxed{x = -2, -1, 1, 2}$$

**8.** End behavior of  $g(x) = 3x^4 - x^3$ .

The leading term is  $3x^4$ . The degree is 4 (even), so both ends travel the same way, and the leading coefficient 3 is positive, so both rise.

$$\text{as } x \rightarrow \infty, g(x) \rightarrow \boxed{+\infty} \qquad \text{as } x \rightarrow -\infty, g(x) \rightarrow \boxed{+\infty}$$

Expected sentence: “Problem 4 had odd degree, so its ends split; here the degree is even, so a negative input raised to an even power is still positive and both ends rise.”

**9.** Least-degree polynomial crossing at  $x = -3$  and touching at  $x = 2$ .

Crossing at  $x = -3$  gives the factor  $(x + 3)$  once; touching at  $x = 2$  gives  $(x - 2)^2$ . With leading coefficient 1:

$$(x + 3)(x - 2)^2 = (x + 3)(x^2 - 4x + 4) = x^3 - 4x^2 + 4x + 3x^2 - 12x + 12$$

Collecting like terms gives the expanded form.

$$x^3 - x^2 - 8x + 12$$

Check the sketch: degree 3 is odd with a positive leading coefficient, so the curve falls on the left and rises on the right, bouncing off the axis at  $x = 2$  on the way up.

**10.** Sketch  $y = x(x + 2)^2(x - 3)$ .

**Zeros and behavior.**  $x = 0$  (multiplicity 1, crosses),  $x = -2$  (multiplicity 2, touches),  $x = 3$  (multiplicity 1, crosses).

$$x = -2, x = 0, x = 3$$

**End behavior.** Degree  $1 + 2 + 1 = 4$  (even) with a positive leading coefficient, so both ends rise.

**Model sketch (manual review — compare the student's curve to this description).**

Coming down from the upper left, the curve touches the axis at  $x = -2$  without crossing, turns back up, then comes down through  $x = 0$ , dips to a minimum between  $x = 0$  and  $x = 3$  (the low point is about  $y = -32$ , at  $x = 2$ ), crosses at  $x = 3$ , and rises steeply. *Credit the sketch when the three intercepts are marked, the bounce at  $x = -2$  is visible, and both ends point up; the vertical scale is the student's own choice.*